ENTHUSIAST ADVANCE COURSE

PHASE : MEB-I (A)

TARGET : PRE-MEDICAL 2024

Test Type : **MINOR** Test Pattern : **NEET (UG)**

TEST DATE : 18-06-2023

ANSWER KEY

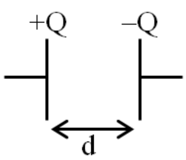
Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	3	1	1	2	3	3	3	1	4	4	3	3	1	3	3	3	2	3	2	3	3	3	3	2	2	2	3	3	4	4
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	3	3	1	2	3	2	2	3	3	2	3	1	1	2	1	4	3	1	4	3	2	3	3	3	3	2	4	2	2	2
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	3	2	2	3	1	1	1	2	2	1	2	3	3	2	4	3	4	3	4	1	3	1	2	4	3	3	4	3	2
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	3	4	4	4	2	3	1	4	2	3	1	2	2	3	1	3	3	1	2	4	1	4	1	2	4	4	2	1	4	3
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	4	4	1	3	3	2	3	1	2	4	1	2	3	4	3	3	2	1	1	3	2	3	3	3	3	2	3	2	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	3	2	2	1	1	2	2	1	3	2	2	3	3	1	2	1	4	2	1	3	1	2	4	3	4	4	3	1	2	2
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	2	3	3	1	1	3	1	1	3	3	3	2	2	4	1	2	3	3	4	3										

HINT – SHEET

SUBJECT : PHYSICS

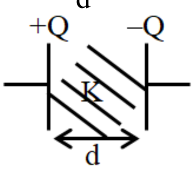
SECTION-A

1. **Ans (3)**



Q = constant

$$C = \frac{\epsilon_0 A}{d}, V_0 = \frac{Q^2}{2C}$$



$$C' = K\frac{\epsilon_0 A}{d} = KC$$

$$U' = \frac{Q^2}{2(KC)} = \frac{1}{K}\frac{Q^2}{2C} = \frac{U_0}{K}$$

2. **Ans (1)**

Both capacitor is connected with battery

V = same

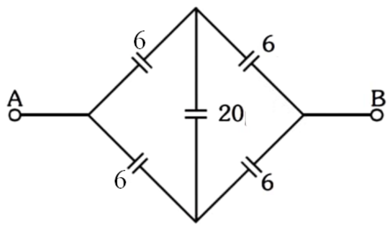
$$C_1 = \frac{\epsilon_0 A}{d}, C_2 = K.\frac{\epsilon_0 A}{d}$$

$$C_2 > C_1 \Rightarrow Q = C\psi$$

$$Q \propto C$$

$\boxed{Q_2 > Q_1}$

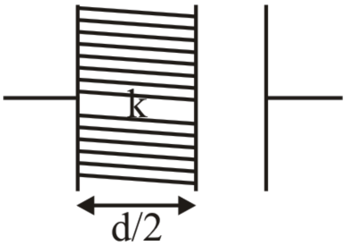
3. **Ans (1)**



This is balanced WSB.

So. $\boxed{C_{eq} = 6\mu F}$

4. Ans (2)

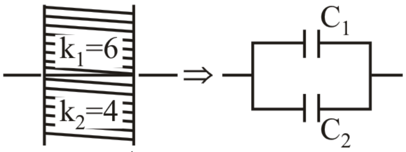


$$C' = \frac{\epsilon_0 A}{\left(d - \frac{d}{2}\right) + \frac{d/2}{k}} = \frac{\epsilon_0 A}{\frac{d}{2}\left(1 + \frac{1}{k}\right)}$$

$$= \frac{2k}{k+1}\frac{\epsilon_0 A}{d}$$

$$C' = \frac{2k}{k+1}C$$

5. Ans (3)

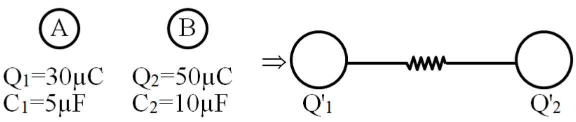


$$C_1 = 6\frac{\epsilon_0 \frac{A}{2}}{d} = \frac{3\epsilon_0 A}{d}$$

$$C_2 = 4\frac{\epsilon_0 \frac{A}{2}}{d} = \frac{2\epsilon_0 A}{d}$$

$$C_{net} = C_1 + C_2 = \frac{5\epsilon_0 A}{d}$$

6. Ans (3)



(1) Common potential (V) $= \dfrac{Q_1 + Q_2}{C_1 + C_2}$

$$V = \frac{30\mu C + 50\mu C}{5\mu F + 10\mu F} = \frac{80}{15}\ \text{Volt}$$

$$V = \frac{16}{3}\ \text{Volt}$$

(2) We know $C = 4\pi\epsilon_0 R \propto R$

$$\frac{Q'_1}{Q'_2} = \frac{R_1}{R_2} = \frac{C_1}{C_2}$$

$$\text{So. } \frac{Q'_1}{Q'_2} = \frac{C_1}{C_2} = \frac{5\mu F}{10\mu F} = \frac{1}{2}$$

7. Ans (3)

$$U = \frac{1}{2}CV^2$$

$$\because C = \frac{\epsilon_0 A}{d}, V = \epsilon d$$

$$U = \frac{1}{2}\left(\frac{\epsilon_0 A}{d}\right)(E^2 d^2) = \frac{1}{2}\epsilon_0 E^2 Ad$$

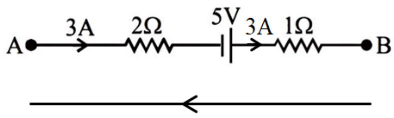
8. Ans (1)

$$\text{Force} = \frac{Q^2}{2A\epsilon_0}$$

$\boxed{F \propto d^0}$

Force is independent of distance between plates.

10. Ans (4)



Apply KVL

$V_B + 3(1) - 5 + 3(2) = V_A$

$V_B - V_A = -4\text{Volt}$

11. Ans (3)

$\tau = PE\sin\theta$

when $\theta = 90° \Rightarrow \tau_{max}$

$\tau_{max.} = PE = (qd)E$

$\tau_{max.} = 1 \times 10^{-6} \times 2 \times 10^{-2} \times 10^5$

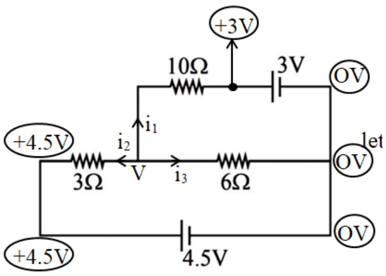
$= 2 \times 10^{-3}$ N–m

12. Ans (3)

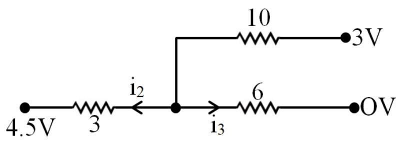
Electrostatic field is conservative so work done is independent of path. it depend on position of points.

So $W_I = W_{II} = W_{III} = W_{IV}$

13. Ans (1)



$\Rightarrow$



Apply KCL $i_1 + i_2 + i_3 = 0$

$$\frac{V-3}{10} + \frac{V-4.5}{3} + \frac{V-D}{6} = 0$$

$$\frac{3V - 9 + 10V - 45 + 5V}{30} = 0$$

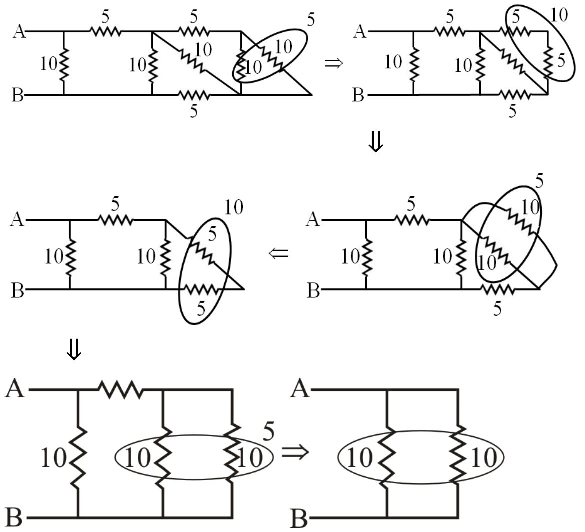
$18V = 54$

$V = 3$

$$\text{So } i_1 = \frac{3-3}{10} = 0$$

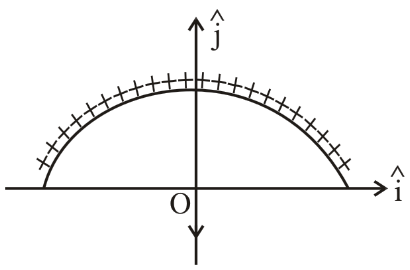
So current in 10Ω is OA.

14. Ans (3)



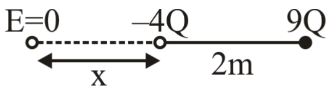
$R_{AB} = 5\Omega$

15. Ans (3)



$$\vec{E} = \frac{2k\lambda}{r}(-\hat{j}) = \frac{2}{4\pi\varepsilon_0}\frac{\lambda}{r}(-\hat{j}) = \frac{\lambda}{2\pi\epsilon_0 r}(-\hat{j})$$

17. Ans (2)



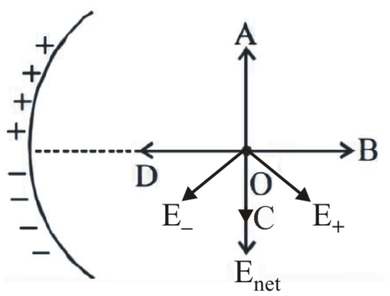
$E_{-4Q} = E_{9Q}$

$$\frac{k(4Q)}{x^2} = \frac{k(9Q)}{(2+x)^2} \Rightarrow \frac{2}{x} = \frac{3}{2+x}$$

$\Rightarrow 4 + 2x = 3x$

$x = 4m$

18. Ans (3)



21. Ans (3)

$$T = 2\pi\sqrt{\frac{\ell}{g}},$$

$T \propto \sqrt{\ell}$

27. Ans (3)

Here, $\mu = m/L = 5.0 \times 10^{-3}$ kg/0.72 m

$= 6.9 \times 10^{-3}$ kg/m

$T = 60$ N

Thus, $v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{60N}{6.9 \times 10^{-3}kg/m}} = 93$ m/s

30. Ans (4)

$$g_{eff} = \frac{g}{\left(1 + \frac{3R}{R}\right)^2} = \frac{g}{16}$$

$mg = 144$

$$mg_{eff} = \frac{144}{g} \times \frac{g}{16} = 9N$$

31. Ans (3)

$$W_d = \frac{1}{4} \times w$$

$$mg_d = \frac{1}{4} \times mg$$

$$g\left(1 - \frac{d}{R}\right) = \frac{g}{4}$$

$$1 - \frac{d}{R} = \frac{1}{4}$$

$$\frac{d}{R} = \frac{3}{4} \Rightarrow d = \frac{3}{4}R$$

32. Ans (3)

$$T^2 = \frac{4\pi^2}{GM}a^3$$

$$T^2 = \frac{4\pi^2}{GM}\left(\frac{r_p + r_A}{2}\right)^3 = \frac{\pi^2}{2GM}(r_p + r_A)^3$$

$mvr =$ constant $\Rightarrow vr =$ constant

$\therefore v_A r_A = v_p r_p$

$\therefore r_A > r_p$

$\therefore v_A < v_p$

33. Ans (1)

A geostationary satellite has a time period of T= 24 hr.

From Kepler's Law,

$$T^2 \propto R^3 \Rightarrow \frac{T_2^2}{T_1^2} = \frac{R_2^3}{R_1^3}$$

$$\Rightarrow \frac{T_2^2}{(24)^2} = \frac{(R + 2.5R)^3}{(R + 6R)^3}$$

$$T = \sqrt{\frac{24 \times 24}{8}} = 6\sqrt{2} \text{ hour}$$

34. **Ans (2)**

$$W = \Delta U = \frac{GMm}{R(R+h)}$$

given, h = 2R

$$W = \frac{GMm}{R(R+2R)} = \frac{GMm}{3R^2}$$

35. **Ans (3)**

$$V_P = V_{Earth} + V_{Moon}$$

$$V_P = \frac{-G(81M)}{D/2} - \frac{G(M)}{D/2}$$

$$= -\frac{162GM}{D} - \frac{2GM}{D} = -\frac{164GM}{D}$$

SECTION-B

36. **Ans (2)**

given $u = 2.1 \times 10^{-9}\ J/m^3$

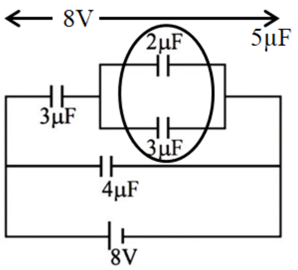
$$u = \frac{1}{2}\epsilon_0 E^2$$

$$E^2 = \frac{2u}{\epsilon_0}$$

$$E = \sqrt{\frac{2u}{\epsilon_0}} = \sqrt{\frac{2 \times 2.1 \times 10^{-19}}{8.85 \times 10^{-12}}}$$

$$\boxed{E = 21.6\frac{N}{C}}$$

37. **Ans (2)**



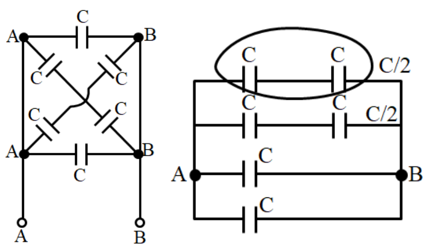
In series $V \propto \frac{1}{C}$

$$V_3 : V_5 = \frac{1}{3} : \frac{1}{5} = 5 : 3$$

$$V_5 = \frac{3}{8} \times 8 = 3V$$

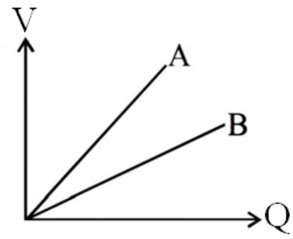
$$Q_{2\mu F} = (2\mu F)(3V) = 6\mu C$$

38. **Ans (3)**



$$C_{AB} = \frac{C}{2} + \frac{C}{2} + C + C = 3C$$

39. **Ans (3)**

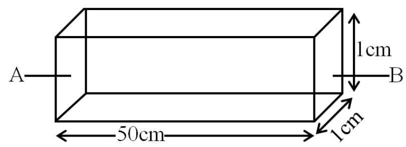


$$\text{Slope} = \frac{V}{Q} = \frac{1}{C}$$

$$(\text{Slope})_A > (\text{Slope})_B$$

$$C_A < C_B$$

40. **Ans (2)**



$$R_{AB} = \frac{\rho\ell}{A}$$

$$\rho = 3.5 \times 10^{-5}\ \Lambda^{-m}$$

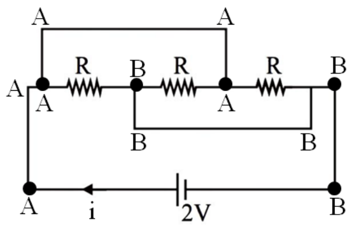
$$\ell = 50cm = 50 \times 10^{-2}m$$

$$A = 1cm \times 1cm = 1\ cm^2 = 10^{-4}m^2$$

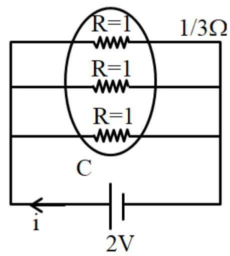
$$R_{AB} = \frac{3.5 \times 10^{-5} \times 50 \times 10^{-2}}{10^{-4}}$$

$$\boxed{R_{AB} = 0.175\Omega}$$

41. **Ans (3)**



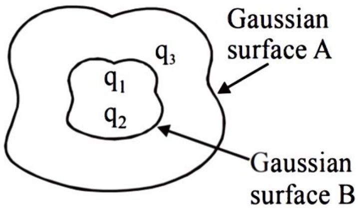
$\Rightarrow$



$$i = \frac{2V}{\frac{1}{3}\Omega}$$

$$i = 6A$$

42. Ans (1)

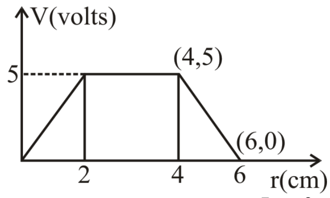


$$\phi_A = \frac{q_1 + q_2 + q_3}{\epsilon_0}$$

$$\phi_A = \frac{-14nC + 78.85nC - 56nC}{\epsilon_0}$$

$$\phi_A = \frac{-8.85nC}{\epsilon_0} = \frac{-8.85 \times 10^{-9}}{8.85 \times 10^{-12}} = 10^{-3}$$

43. Ans (1)



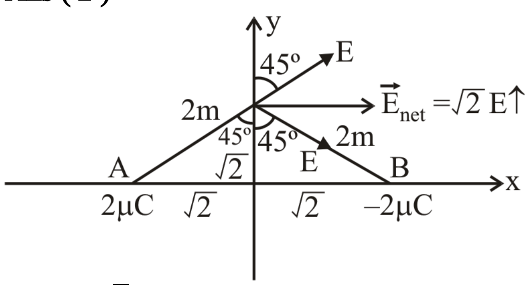
$$\text{Slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{5 - 0}{4 - 6} = -2.5$$

$$E = \frac{-dv}{dr} = -\,(\text{slope})$$

$$E = -(-\,2.5)$$

$$E = 2.5 \text{ V/cm}$$

44. Ans (2)



$$E = \frac{k(\sqrt{2}\mu C)}{(2)^2} = \frac{9 \times 10^9 \times \sqrt{2} \times 10^{-6}}{(2)^2}$$

$$E_{net} = \frac{\sqrt{2} \times 9 \times 10^9 \times \sqrt{2} \times 10^{-6}}{2 \times 2} = \frac{9000}{2}\hat{i}$$

46. Ans (4)

The equation can be written as

$$y = 4\sin\left(4\pi t - \frac{\pi x}{16}\right)$$

Wave velocity is given by

$$v = \frac{\text{coefficient of } t\,(\omega)}{\text{coefficient of } x\,(k)}$$

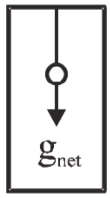
$$\Rightarrow v = \frac{4\pi}{\frac{\pi}{16}} = 64\,\frac{cm}{s} \text{ in } +\text{ve}\,x - \text{direction.}$$

47. Ans (3)

$$\frac{V_{H_2}}{V_{He}} = \sqrt{\frac{r_{H_2}}{r_{He}} \times \frac{M_{He}}{M_{H_2}}}$$

$$\frac{V_{H_2}}{V_{He}} = \sqrt{\frac{7}{5} \times \frac{3}{5} \times \frac{4}{2}} = \frac{\sqrt{42}}{5}$$

48. Ans (1)



$$\uparrow y = kt^2, \quad V = \frac{dy}{dt} = 2kt$$

$$a = \frac{dy}{dt} = 2k = 2(1) = 2 \text{ m/s}^2$$

$$g_{net} = g + a$$

$$g_{net} = 10 + 2 = 12 \text{ m/s}^2$$

$$\text{Now } \frac{T_1}{T_2} = \sqrt{\frac{g_{net}}{g}} = \sqrt{\frac{12}{10}} = \sqrt{\frac{6}{5}}$$

49. Ans (4)

$$V_e = \sqrt{\frac{2GM}{R}}$$

$$\frac{(V_e)_A}{(V_e)_B} = \sqrt{\left(\frac{M_A}{M_B}\right)\left(\frac{R_B}{R_A}\right)} = \sqrt{\left(\frac{4M}{M}\right)\left(\frac{2R}{R}\right)}$$

$$= \sqrt{\frac{8}{1}} = \frac{2\sqrt{2}}{1}$$

50. Ans (3)

$$U = -\frac{GMm}{r} \qquad T = -\frac{GMm}{2r}$$

$$K = \frac{GMm}{2r}$$

For A For B

$$U_A = -\frac{GMm}{4R} \quad U_B = -\frac{GMm}{6R} \quad \frac{U_A}{U_B} = \frac{6}{4} = \frac{3}{2}$$

$$K_A = \frac{GMm}{8R} \quad K_B = \frac{GMm}{12R} \quad \frac{K_A}{K_B} = \frac{12}{8} = \frac{3}{2}$$

$$T_A = -\frac{GMm}{8R} \quad T_B = -\frac{GMm}{12R} \quad \frac{T_A}{T_B} = \frac{12}{8} = \frac{3}{2}$$

SUBJECT : CHEMISTRY

SECTION-A

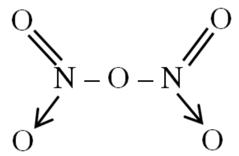
52. Ans (3)

Fact

56. Ans (2)

UUT ⇒ 113

57. **Ans (4)**



59. **Ans (2)**

$P + Cl_2 \longrightarrow PCl_3$ (A)

$PCl_3 \xrightarrow{H_2O} H_3PO_3$

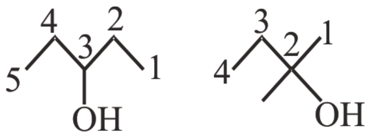
P + excess $Cl_2 \longrightarrow PCl_5$(B)

$PCl_5 \xrightarrow{H_2O} H_3PO_4$

62. **Ans (3)**

B.E $Cl_2 > Br_2 > F_2 > I_2$

73. **Ans (3)**



Chain isomers

SECTION-B

86. **Ans (3)**

Acidic nature = $H_3PO_2 > H_3PO_3$

98. **Ans (4)**

Functional group —O— ether is absent.

SUBJECT : BIOLOGY-I

SECTION-A

104. **Ans (3)**

NCERT : 26

111. **Ans (1)**

NCERT XII Pg # 24

112. **Ans (4)**

NCERT Pg. # 26

117. **Ans (2)**

NCERT XII Pg. # 12, 31

Papaya, date palm & maize have unisexual flowers

118. **Ans (1)**

NCERT-XII, Pg.#28

120. **Ans (3)**

Module page number 170

125. **Ans (3)**

NCERT (XIIth) Pg. # 76

129. **Ans (1)**

NCERT (XIIth) Pg. # 74

132. **Ans (1)**

NCERT-XII, Pg. # 76, 77

134. **Ans (3)**

NCERT XIIth, Pg.No # 77

SECTION-B

144. **Ans (3)**

NCERT XII Pg # 33

According to the NCERT, removal of anthers from the flower bud before the anther dehisces is called emasculation.

In this question word **'during'** is given in place of word **'before'** in statement I, which is not correct.

145. **Ans (3)**

NCERT XII page 34-35, C & D are true

146. **Ans (3)**

NCERT–XII Pg. # 75

149. **Ans (2)**

NCERT Pg. # 89

SUBJECT : BIOLOGY-II

SECTION-A

155. **Ans (1)**

NCERT (XII) Pg# 165 Para: 9.1

156. **Ans (2)**

NCERT (XII) Pg# 169 Para: 9.1.2

157. **Ans (2)**

NCERT-XII Pg# 167 Para: 9.1.2

158. **Ans (1)**

NCERT-XII Pg# 169 Para: 9.1.4

159. **Ans (3)**

NCERT XIIth Pg # 169

160. **Ans (2)**

NCERT-XII Pg. # 167

161. Ans (2)

NCERT (XIth) Pg. # 102

164. Ans (1)

NCERT-XI-104

178. Ans (1)

NCERT Pg. # 228

180. Ans (2)

NCERT XII Pg # 232

SECTION-B

188. Ans (1)

NCERT (XII) Pg# 168 Para: 9.1.2

189. Ans (3)

NCERT (XII) Pg# 168 Para: 9.1.2

190. Ans (3)

NCERT-XII Pg# 167 Para: 9.1.2

192. Ans (2)

NCERT-XII, Pg. # 102

194. Ans (4)

NCERT Pg# 103, fig.-7.4(b)